

WILLIAMSBURG JAMESTOWN, VA, USA

WMO: 720297

Lat: 37.239N

Lon: 76.716W

Elev: 15

StdP: 101.15

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03730

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -4.1	(d) -17.1	(e) 0.8	(f) -4.8	(g) -13.8	(h) 1.1	(i) -1.9	(j) 10.4	(k) 8.6	(l) 8.9	(m) 7.5	(n) 2.1	(o) 340	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	36.3	25.8	34.2	24.8	33.0	24.4	27.4	32.9	26.6	32.1	25.9	31.1	2.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.7	25.1	20.3	28.9	24.1	19.0	28.0	86.0	32.1	82.6	31.8	80.2	31.2	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 7.0	(c) 5.6	(e) -10.5	(f) 38.6	(g) 3.0	(h) 1.5	(i) -12.6	(j) 39.7	(k) -14.4	(l) 40.6	(m) -16.1	(n) 41.5	(o) -18.2	(p) 42.6		
(a) 8.1	(b) 7.0	(c) 5.6	(e) -11.5	(f) 28.4	(g) 2.7	(h) 1.0	(i) -13.5	(j) 29.1	(k) -15.1	(l) 29.7	(m) -16.6	(n) 30.2	(o) -18.6	(p) 30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	5.2	6.8	10.6	16.6	21.5	25.8	27.8	27.0	23.8	17.8	11.3	7.8		
	DBStd	9.12	5.76	5.58	5.57	4.80	4.30	3.13	2.48	2.45	3.23	4.47	4.58	5.04		
	HDD10.0	499	170	121	62	4	0	0	0	0	0	2	38	103		
	HDD18.3	1695	408	326	247	88	20	0	0	0	1	64	214	327		
	CDD10.0	3017	21	30	81	203	357	473	553	527	415	244	77	36		
	CDD18.3	1171	0	1	7	38	118	223	294	268	166	49	4	2		
	CDH23.3	11228	1	13	72	364	1015	2223	3222	2657	1311	327	18	6		
	CDH26.7	4816	0	1	12	116	387	998	1552	1185	486	78	1	0		
(14) Wind	WSAvg	2.1	2.5	2.7	2.8	2.7	2.0	1.8	1.6	1.4	1.5	1.9	2.0	2.1		
(15) Precipitation	PrecAvg	1224	99	86	109	82	110	96	141	129	116	85	82	87		
	PrecMax	1921	207	206	236	166	238	255	304	351	542	254	202	169		
	PrecMin	752	16	20	5	17	22	16	16	32	16	1	10	10		
	PrecStd	254	42	41	47	36	52	51	66	80	105	56	49	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	25.9	28.0	32.5	34.1	37.6	38.7	37.3	35.8	32.1	25.3	23.9		
		MCWB	17.2	18.6	18.4	20.8	23.8	25.3	25.9	26.4	24.5	23.5	18.4	19.4		
	2%	DB	19.8	22.2	25.9	29.9	32.5	35.9	36.3	35.1	32.9	28.8	22.9	21.3		
		MCWB	15.7	15.7	16.8	19.7	23.0	25.0	26.4	25.4	23.9	22.5	17.3	18.4		
	5%	DB	17.3	18.9	22.6	27.5	31.1	33.0	34.9	33.7	32.0	27.1	21.3	18.0		
		MCWB	13.7	14.7	15.3	18.4	22.7	24.1	25.6	24.8	23.8	21.2	17.1	15.8		
	10%	DB	14.1	16.2	20.0	25.0	28.7	32.2	32.9	32.4	29.1	25.0	18.9	16.2		
		MCWB	10.8	11.4	14.4	17.9	21.6	23.9	25.0	24.6	22.4	20.2	15.5	13.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.4	20.1	23.3	25.3	27.3	28.9	28.2	26.3	25.3	20.9	21.1		
		MCDB	21.0	23.6	25.5	27.5	32.1	33.7	35.5	33.3	32.1	28.0	23.3	22.7		
	2%	WB	16.6	17.5	18.5	21.5	24.2	26.2	27.6	27.2	25.6	24.0	19.3	18.4		
		MCDB	19.3	19.7	22.5	26.9	30.1	32.9	33.6	32.0	30.6	26.8	21.8	20.1		
	5%	WB	14.2	15.1	17.1	20.1	23.5	25.5	26.8	26.4	25.0	22.9	17.8	16.7		
		MCDB	16.8	18.6	21.2	25.4	28.4	31.6	32.5	31.1	29.3	25.6	19.8	18.1		
	10%	WB	11.8	12.1	15.3	18.8	22.7	24.8	26.1	25.7	24.2	21.2	16.3	13.9		
		MCDB	13.8	15.2	18.9	23.4	27.0	30.1	31.4	30.3	27.7	24.0	18.8	15.8		
(35) Mean Daily Temperature Range	MDBR		10.0	11.0	11.6	12.1	11.0	10.6	10.3	9.9	10.1	10.5	10.9	10.1		
	5% DB	MCDBR	13.4	14.3	14.4	14.1	12.6	11.8	11.6	11.1	11.6	11.6	12.7	11.8		
		MCWBR	10.2	9.6	7.8	5.9	5.3	4.3	3.9	3.7	4.1	5.6	7.8	9.3		
	5% WB	MCDBR	12.5	13.9	12.5	11.8	10.8	10.6	10.4	9.7	9.3	9.1	10.6	11.0		
		MCWBR	10.8	10.5	8.4	6.1	5.0	4.3	4.1	3.9	3.9	5.0	8.1	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.316	0.345	0.391	0.432	0.486	0.515	0.511	0.408	0.352	0.321	0.308		
	taud		2.537	2.496	2.448	2.329	2.245	2.143	2.079	2.078	2.352	2.507	2.546	2.559		
	Ebn at Noon		892	921	920	890	855	806	778	771	842	868	863	863		
	Edh at Noon		77	91	105	125	138	153	162	159	114	89	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.33	3.08	4.10	5.19	5.87	6.34	6.04	5.46	4.43	3.50	2.63	2.07		
	RadStd		0.19	0.34	0.33	0.39	0.53	0.43	0.30	0.28	0.40	0.41	0.22	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+106

Nomenclature: See separate page